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Analytical solution for Kelly's criterion for multiple outcomes

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We present an elementary analytical representation of the optimal portfolio in multi-outcome prediction markets, extending the well-known Kelly criterion from the binary model. Specifically, the solution is to invest a fraction $\mathbb{P}^M(\omega)$ of one's wealth in each market outcome ω, where $\mathbb{P}^M$ represents the agent's subjective (physical) measure and $\mathbb{P}^Y$ the quoted (risk-neutral) measure. The resulting final portfolio payoff is represented by the likelihood ratio $\frac{\mathbb{P}^M}{\mathbb{P}^Y}$. This ratio is shown to be optimal for logarithmic utility, so once identified, no further calculations are required. Moreover, its expected logarithmic return corresponds to the relative entropy (Kullback–Leibler divergence), offering an intriguing connection to information theory. This framework also accommodates trading constraints: in such cases, the agent must find the replicable portfolio V, associated with a state price density $\mathbb{P}^V$, that is closest to $\mathbb{P}^M$ in the sense of relative entropy.

Keywords: Optimal portfolio; Kelly criterion; Prediction markets; Logarithmic utility; Relative entropy; Risk-neutral measure; Physical measure

JEL Classification: G11, G13, D81

1. Introduction

Kelly's criterion, introduced by Kelly (1956), prescribes how to maximize the long-term growth rate of wealth in repeated bets. It invests capital so as to maximize the expected logarithm of final wealth, thereby balancing risk of ruin against high returns. In the simplest (binary) case, the Kelly formula depends on the investor's subjective odds compared to the quoted market odds (Thorp 1971, MacLean *et al.* 2011).

Subsequent work explored more general settings. Breiman (1961) provided an early extension to repeated investing under favorable games. In continuous-time, it is well known that a discrete Kelly framework can converge to the classical Merton portfolio problem (Merton 1971), and indeed these links have been studied in, e.g. Browne and Whitt (1996).

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In this paper, we further develop these connections by giving an *elementary* analytical solution for multi-outcome prediction markets. To the best of our knowledge, previous general approaches—such as the recursive method of Smoczynski and Tomkins (2010), the convex optimization of Busseti *et al.* (2016), or the utility-based analyses of Whelan (2025)—did not explicitly recognize that the logarithmically optimal (Kelly) solution for multiple outcomes arises from the ratio of the investor's physical measure $\mathbb{P}^M$ to the market's risk-neutral measure $\mathbb{P}^Y$. Once this ratio is identified, the Kelly portfolio is found immediately.

From a financial perspective, the expected logarithmic return of this ratio equals the relative entropy between $\mathbb{P}^M$ and $\mathbb{P}^Y$. This is not a minor technical detail but a fundamental structural insight: relative entropy (or Kullback–Leibler divergence) quantifies how far the investor's subjective probabilities deviate from the market's risk-neutral probabilities, and thus measures the value of private information in precise statistical terms. Historically, this connection is deeply rooted. Claude Shannon, the founder of information theory, introduced the concept of entropy in his landmark 1948 paper (Shannon 1948), and Kelly, who worked closely with Shannon at Bell Labs, built directly on this framework. In fact, Kelly's 1956 paper explicitly described the criterion in terms of an *information rate*, which is mathematically identical to relative entropy. In this sense, the Kelly criterion sits at the intersection of finance and information theory from its very inception. Highlighting this perspective clarifies why the Kelly solution has a universal character: it is simultaneously the growth-optimal portfolio and the information-theoretic likelihood ratio. In a constrained setting (e.g. when perfect replication of this ratio is not possible), the optimal payoff becomes the state-price density $\mathbb{P}^V$ closest to $\mathbb{P}^M$ in the Kullback–Leibler sense—an appealing viewpoint for unifying ideas in finance, probability, and statistics. For a standard modern reference on entropy, relative entropy, and their role in information theory, see Cover and Thomas (2006).

Moreover, our analysis clarifies an important structural distinction. While the terminal wealth that maximizes expected utility is uniquely defined, the *replication strategy* implementing this payoff is not unique because of the linear dependence of Arrow–Debreu securities. The most natural representation is the *full allocation* in Arrow–Debreu securities, a perspective that mirrors the statistical interpretation in Shafer (2021). In contrast, approaches that emphasize *minimal exposure* allocations are more difficult to characterize, since they require additional reference to the numéraire $\mathbb{P}^Y$. Our contribution is to show that even these more complicated allocations can be deduced systematically from the full-investment solution. In particular, for general CRRA preferences we obtain explicit analytical weights, thereby providing closed-form solutions that have proved elusive in the previous literature.

Since the fraction $\mathbb{P}^M(\omega) := p(\omega|M)$ that the agent invests on scenario ω does not depend on the chosen benchmark asset, the solution is *numéraire-invariant*. However, when certain scenarios are assigned very small $\mathbb{P}^M$-to-$\mathbb{P}^Y$ ratios, the resulting payoff can be small and unfavorable in those states; risk-averse investors often dampen such exposures by choosing an alternative distribution $\mathbb{P}^V$ that lies between $\mathbb{P}^M$ and $\mathbb{P}^Y$. In effect, they sacrifice some expected growth for reduced risk—a procedure foreign to pure *statistical* likelihood but standard in finance.

1.1. Organization

Section 2 introduces the Arrow–Debreu framework for pricing discrete outcomes, defines the risk-neutral measure $\mathbb{P}^Y$, and shows how an investor's subjective belief $\mathbb{P}^M$ induces a payoff in the form of their likelihood ratio relative to $\mathbb{P}^Y$. Section 3 develops this result as the *multi-outcome Kelly strategy*, illustrating its interpretation in binary and multi-outcome examples and explaining how risk aversion modifies the pure likelihood ratio into tempered measures $\mathbb{P}^V$. Section 4 explores the replication of these payoffs in Arrow–Debreu securities, highlighting the non-uniqueness of hedging strategies and distinguishing between full-investment and minimal-exposure allocations. Section 5 establishes log-utility optimality of the likelihood ratio, links expected log returns to relative entropy, and generalizes the framework to CRRA utilities. Section 6 extends the analysis to financial markets where state price densities are fixed by traded securities rather than individual outcomes, and derives the growth-optimal portfolio as the distribution closest to $\mathbb{P}^M$ in relative entropy. Section 7 shows how a mean–variance approximation recovers the Kelly allocation as a second-order expansion. Section 8 discusses implications for risk management and the interpretation of 'good' and 'bad' properties of the Kelly strategy in light of likelihood modeling. Finally, section 9 concludes.

2. Prices as likelihood ratios

2.1. Risk-neutral measure

Consider a finite set of possible outcomes $\Omega = \{\omega_1, \omega_2, \ldots, \omega_n\}$. A prediction market 'quotes' each outcome by giving the fraction $p(\omega|Y)$ of a reference asset Y to invest at $t = 0$ in order to receive one unit of Y at $t = T$ if scenario ω occurs. Formally:

$$p(\omega|Y)Y[0] \longrightarrow Y[T]\mathbf{1}_{\{\omega\}}, \tag{1}$$

where $\mathbf{1}_{\{\omega\}}$ is the indicator function taking value 1 if ω happens and 0 otherwise. These digital contracts are known as Arrow–Debreu securities (Arrow and Debreu 1954). Because

$$\sum_{\omega \in \Omega} p(\omega|Y) = 1,$$

the weights $\{p(\omega|Y)\}$ can be interpreted as probabilities, often called the *risk-neutral* measure $\mathbb{P}^Y$. Equivalently, the reciprocal $\frac{1}{p(\omega|Y)}$ can be viewed as the 'odds' for outcome ω. The conversion from probabilities to odds implicitly assumes $p(\omega|Y) > 0$ (the market does not trade outcomes deemed impossible).

Since Y is settled at $t = T$, each Arrow–Debreu security is already 'discounted,' effectively replacing currency by a zero-coupon bond if needed. Thus, no further discounting is required.

2.2. Physical measure

From standard finance theory, we know there is no guarantee that $\mathbb{P}^Y$ matches the true frequencies of future scenarios.

Instead, these true frequencies should be reflected by the *physical measure* $\mathbb{P}^M$. While the classical view treats this measure as objectively given, consistent with the probabilistic approach, we adopt a statistical perspective in which the agent subjectively estimates her own $\mathbb{P}^M$.

This is conceptually different from traditional financial thinking. Once the agent specifies $\mathbb{P}^M$, the resulting payoff is uniquely determined: she invests a fraction $p(\omega|M)$ of her wealth in each outcome ω, exactly matching her belief. This investment rule emerges directly from the agent's subjective model and is independent of the market's quoted odds $\mathbb{P}^Y$. The strategy is therefore numéraire-invariant and requires no further optimization. Remarkably, the resulting payoff turns out to maximize logarithmic utility—this is a derived property, not an assumed objective.

By analogy with the risk-neutral construction, we may interpret the agent's investment as assigning a hypothetical state price under $\mathbb{P}^M$:

$$p(\omega|M)M[0] \longrightarrow M[T]\mathbf{1}_{\{\omega\}}, \tag{2}$$

where M is the agent's fund. This compact representation reinforces the idea that her wealth allocation mirrors her own beliefs.

In practice, agents often temper their subjective views by blending $\mathbb{P}^M$ with $\mathbb{P}^Y$, yielding an intermediate measure $\mathbb{P}^V$. This updated model—typically more conservative in adverse scenarios—results in a payoff V that is less exposed to downside risk. From a statistical standpoint, this choice corresponds to adopting a different model $\mathbb{P}^V$, which is ultimately what the market observes through the revealed payoff.

Traditional portfolio optimization methods can thus be reinterpreted: rather than searching for an optimal portfolio directly, they can be viewed as procedures for identifying a suitable candidate measure $\mathbb{P}^V$. Once this measure is specified, the corresponding payoff is uniquely determined as the log-optimal portfolio under $\mathbb{P}^V$, and no further optimization is needed. The agent's decision problem is therefore reduced to selecting the measure to communicate to the market, not the portfolio itself.

2.3. Prices as likelihood ratios

To prevent arbitrage, two equivalent ways of investing in scenario ω must deliver the same payoff. Specifically:

(i) Invest $M[0]$ directly to receive $\frac{1}{p(\omega|M)}M[T]$ in scenario ω.

(ii) Convert M into the reference asset Y at time zero by purchasing $M_Y(0)$ units of Y. This yields $\frac{M_Y(0)\,Y[T]}{p(\omega|Y)}$ if ω occurs.

2.3.1. Notation. $M_Y(t)$ denotes the number of units of asset Y required at time t to acquire one unit of asset M. Equivalently,

$$M[t] = M_Y(t)\,Y[t],$$

where $M[t]$ and $Y[t]$ denote the assets M and Y considered at time t (i.e. the assets stamped at time t). Throughout, parentheses $(\cdot)$ denote numerical arguments (such as indices or parameters), and square brackets $[\cdot]$ denote assets indexed by time.

Equating the two payoffs in every scenario ω gives

$$\frac{1}{p(\omega|M)}M[T]\,\mathbf{1}_{\{\omega\}} = \frac{M_Y(0)\,Y[T]}{p(\omega|Y)}\,\mathbf{1}_{\{\omega\}},$$

which rearranges to the well-known *likelihood ratio* form:

THEOREM 2.1 Prices as Likelihood Ratios *Let $\mathbb{P}^M$ be the physical measure and $\mathbb{P}^Y$ be the risk-neutral measure. Then the price of the investor's fund M follows*

$$M_Y(T,\omega) = \frac{p(\omega|M)}{p(\omega|Y)} \times M_Y(0). \tag{3}$$

In other words, the agent invests $p(\omega|M) \times M_Y(0)$ units of capital to scenario ω, *receiving odds* $\frac{1}{p(\omega|Y)}$.

2.3.2. Martingale property. Under the risk-neutral measure $\mathbb{P}^Y$, $M_Y(t)$ is a martingale:

$$\mathbb{E}^Y[M_Y(T)] = \int_\Omega \frac{p(\omega|M)}{p(\omega|Y)} M_Y(0)\,p(\omega|Y)\,\mathrm{d}\omega = M_Y(0).$$

2.3.3. Implied measure via payoff. Equation (3) implies that *any* payoff $V_Y(T)$ can be associated with a unique state price density $\mathbb{P}^V$, given by:

$$p(\omega|V) = \frac{V_Y(T,\omega)}{V_Y(0)} \times p(\omega|Y). \tag{4}$$

The agent can instead choose a payoff $V_Y(T)$ different from the payoff $M_Y(T)$ implied by the original physical state price density $\mathbb{P}^M$; the chosen payoff $V_Y(T)$ then implies a different measure $\mathbb{P}^V$ via equation (4). The fact that one can infer the state price density from the payoff is known as the change of numeraire in option pricing, introduced by Geman *et al.* (1995).

EXAMPLE 2.2 Roulette Suppose an investor with total wealth $V_Y(0) = 100$ places \$1 on 'red' in a fair roulette game. Final wealth is 101 if red occurs, and 99 otherwise. Since $\mathbb{P}^Y(\omega) = 1/2$ for red/black, the implied state price density is

$$p(\omega|V) = \begin{cases} \frac{101}{100} \times \frac{1}{2} = 0.505, & \omega = \text{Red}, \\ \frac{99}{100} \times \frac{1}{2} = 0.495, & \omega = \text{Black}. \end{cases}$$

By choosing this bet, the investor 'shifts' red's effective probability from 0.5 to 0.505.

Although the investor's $\mathbb{P}^M$ is never directly observed by the market, her chosen payoff does induce an *implied* measure $\mathbb{P}^V$. The only way the market can validate an investor's beliefs is by observing that her realized payoffs systematically outperform or underperform the risk-neutral odds over time, akin to a likelihood ratio test.

3. Kelly's criterion as the likelihood ratio

Theorem 2.1 already shows how an investor's distributional view $\mathbb{P}^M$ translates into a *payoff* given by

$$\frac{p(\omega|M)}{p(\omega|Y)}.$$

Concretely, the investor allocates $p(\omega|M)$ fraction of her total wealth $M_Y(0)$ to the Arrow–Debreu security on scenario ω, receiving odds $\frac{1}{p(\omega|Y)}$. This *unilateral* approach does not adjust the bet size depending on the quoted market odds; it is fully determined by $\mathbb{P}^M$. From another perspective, the payoff

$$M_Y(1,\omega) = \frac{p(\omega|M)}{p(\omega|Y)} M_Y(0)$$

is precisely the discrete-time analogue of the 'growth-optimal portfolio.'

3.1. Binary case and the Kelly fraction

As a simple illustration, consider the classic binary outcome $\Omega = \{H, T\}$ with $\mathbb{P}^M = (p, 1-p)$ and $\mathbb{P}^Y = (q, 1-q)$. Then

$$M_Y(1,H) = \frac{p}{q} M_Y(0), \quad M_Y(1,T) = \frac{1-p}{1-q} M_Y(0).$$

Equivalently, the investor allocates a fraction p to scenario H and fraction $(1-p)$ to scenario T, independently of the market odds $\frac{1}{q}$. If $p > q$, the investor has a perceived 'edge' on outcome H, and scenario T may yield a final wealth smaller than $M_Y(0)$. This model corresponds to a binomial model with uptick $u = \frac{p}{q}$ and downtick $d = \frac{1-p}{1-q}$. The fraction of wealth lost in scenario T is

$$1 - \frac{1-p}{1-q} = \frac{p-q}{1-q},$$

often called the *Kelly fraction*, i.e. the fraction of the investor's wealth lost on the losing outcome.

The likelihood approach provides immediately a solution for multiple outcomes that have not been covered in the previous literature. Let us illustrate this concept on the set of three possible outcomes, corresponding to a soccer game.

Example 3.1 Soccer Betting Let $\Omega = \{\text{Win}, \text{Draw}, \text{Loss}\}$. Market odds are $\left(\frac{1}{q^W}, \frac{1}{q^D}, \frac{1}{q^L}\right)$, i.e. $\mathbb{P}^Y = (q^W, q^D, q^L)$. Suppose an investor believes $\mathbb{P}^M = (p^W, p^D, p^L)$. By Kelly's multi-outcome principle, she allocates fractions p^W, p^D, p^L of her wealth to the respective Arrow–Debreu securities, receiving odds $\frac{1}{q^W}, \frac{1}{q^D}, \frac{1}{q^L}$. The resulting payoff is

$$M_Y(1,\omega) = \begin{cases} \dfrac{p^W}{q^W} M_Y(0), & \omega = \text{Win}, \\ \dfrac{p^D}{q^D} M_Y(0), & \omega = \text{Draw}, \\ \dfrac{p^L}{q^L} M_Y(0), & \omega = \text{Loss}. \end{cases}$$

If $p^L < q^L$, the payoff under Loss is below $M_Y(0)$. An investor disliking that shortfall can *temper* her original $\mathbb{P}^M$ toward a new distribution $\mathbb{P}^V$, trading away some expected growth for risk reduction. Numerically, suppose $\mathbb{P}^Y = (0.5, 0.3, 0.2)$ and $\mathbb{P}^M = (0.6, 0.3, 0.1)$. Then investing the fractions of the agent's wealth $p^W = 0.6, p^D = 0.3, p^L = 0.1$ yields payoffs

$$\begin{aligned}(M_Y(1,\text{Win}), M_Y(1,\text{Draw}), M_Y(1,\text{Loss})) \\ = (1200, 1000, 500),\end{aligned} \quad (5)$$

for an initial capital $M_Y(0) = 1000$. Note that the allocation of 10% of the wealth to the Arrow–Debreu security corresponding to the outcome 'Loss' may seem counterintuitive, because its expected simple return is -0.5. Nonetheless, from the viewpoint of both log utility maximization and mean–variance optimization, it remains optimal, as explained in the following text. The key reason is that Arrow–Debreu securities are negatively correlated (if one outcome occurs, the others do not), so the resulting portfolio's variance reduction outweighs this asset's negative expected return.

If a payoff of \$500 under Loss seems too low, it can be raised to \$800 by selecting a new portfolio with cost

$$\begin{aligned}V_Y(0) = \mathbb{E}^Y[V_Y(1)] = 0.5 \times 1200 + 0.3 \\ \times 1000 + 0.2 \times 800 = 1060,\end{aligned}$$

and payoff

$$(V_Y(1,\text{Win}), V_Y(1,\text{Draw}), V_Y(1,\text{Loss})) = (1200, 1000, 800),$$

thus inducing a new implied measure $\mathbb{P}^V = (0.56604, 0.28302, 0.15094)$ per equation (4). This move from $\mathbb{P}^M$ toward $\mathbb{P}^Y$ illustrates standard risk aversion.

4. Hedging decomposition in Arrow–Debreu securities

The price representation from equation (3),

$$M_Y(1,\omega) = \frac{p(\omega|M)}{p(\omega|Y)} M_Y(0),$$

is multiplicative. While it is realized by placing $p(\omega|M)\, M_Y(0)$ units of the agent's wealth on outcome ω at market odds $1/p(\omega|Y)$, it is instructive to obtain the identical representation in additive form using the profit–loss distribution from individual bets. For this reason, consider the set of Arrow–Debreu securities $X^1, \ldots, X^N$ with payoffs

$$X_Y^i(T,\omega) = \begin{cases} 1, & \omega = \omega_i, \\ 0, & \omega \neq \omega_i. \end{cases}$$

At time $t = 0$, the market prices each contract as $X^i(0) = \mathbb{E}^Y[X_Y^i(T)] = p(\omega_i|Y)$. The set of Arrow–Debreu securities is linearly dependent, which follows from the elementary fact that

$$\sum_{i=1}^{N} X_Y^i(t) = 1 \quad \text{for all } t.$$

4.1. Additive representation: absolute positions

The agent buys Δ^i units of X^i, creating a portfolio V,

$$V_Y = \sum_{i=1}^{N} \Delta^i X_Y^i.$$

For a static portfolio (constant Δ^i), in scenario ω_j the portfolio realizes the payoff

$$V_Y(T, \omega_j) = \Delta^j.$$

Evaluating at $t = 0$,

$$\begin{aligned} V_Y(0) &= \sum_{i=1}^{N} V_Y(T, \omega_i)\, X_Y^i(0) \\ &= \sum_{i=1}^{N} V_Y(T, \omega_i)\, p(\omega_i|Y) = \mathbb{E}^Y[V_Y(T)], \end{aligned}$$

confirming the martingale property.

The wealth dynamics are

$$V_Y(T, \omega) - V_Y(0) = \sum_{i=1}^{N} \Delta^i \left(X_Y^i(T, \omega) - X_Y^i(0) \right),$$

which yields the additive representation

$$V_Y(T, \omega_j) = V_Y(0) + \Delta^j - \sum_{i=1}^{N} \Delta^i X_Y^i(0). \tag{6}$$

4.2. Additive representation: relative positions

Another representation is in relative terms: let w^i denote the fraction of wealth invested in X^i. Then

$$\frac{V_Y(t)}{V_Y(0)} = \sum_{i=1}^{N} w^i \frac{X_Y^i(t)}{X^i(0)}, \quad w^i = \Delta^i \frac{X^i(0)}{V_Y(0)}.$$

Rewriting (6) in relative terms and using $X^i(0) = p(\omega_i|Y)$, we obtain

$$\begin{aligned} V_Y(T, \omega_j) &= V_Y(0) + \frac{w^j}{p(\omega_j|Y)} V_Y(0) - \left(\sum_{i=1}^{N} w^i \right) V_Y(0) \\ &= V_Y(0) \left(1 + \frac{w^j}{p(\omega_j|Y)} - \sum_{i=1}^{N} w^i \right). \end{aligned}$$

4.3. Link to utility maximization

The traditional approach in the literature is to maximize the expected utility of the scaled final wealth,

$$\begin{aligned} &\max_{V_Y(T)} \mathbb{E}^M\, U\left(\frac{V_Y(T)}{V_Y(0)} \right) \\ &\quad = \max_{w^i} \sum_{j=1}^{N} U\left(1 + \frac{w^j}{p(\omega_j|Y)} - \sum_{i=1}^{N} w^i \right) p(\omega_j|M), \end{aligned}$$

for a concave increasing utility U. The formulation on the left admits a unique solution (see Kramkov and Schachermayer 1999, Karatzas and Shreve 1998), while the formulation on the right highlights the replication in Arrow–Debreu securities. Because Arrow–Debreu securities are linearly dependent, however, the optimal set of weights w^i is not unique, which complicates the analysis.

Two corner cases illustrate the alternatives:

- **Full investment:** imposing $\sum_{i=1}^{N} w^i = 1$ corresponds to being fully invested in Arrow–Debreu securities. This case is the most natural: it connects directly with the statistical formulation of betting (cf. Shafer 2021), it yields a *unilateral* allocation depending only on $\mathbb{P}^M$, and it reflects the optimal split of wealth for an agent who does not face transaction costs (such as a bookmaker accepting bets on all outcomes and thus necessarily exposed to all scenarios). In related work, Vecer (2019) documented how a market maker can dynamically update the odds so that the resulting positions align with the market maker's intended frequencies across all outcomes. Here, the relative weights reduce to $w^i = p(\omega_i|M)$, and the solution coincides with the classical Kelly criterion.
- **Minimal exposure:** alternatively, one may seek to minimize total exposure (e.g. to account for transaction costs). This formulation depends non-trivially on the market measure $\mathbb{P}^Y$, so the resulting allocation changes if the quoted odds change, even if $\mathbb{P}^M$ remains the same. The minimal-exposure allocation is thus not a native statement of the problem but rather a construction derived from the full-investment solution. It lacks the direct statistical interpretation of placing stakes proportional to subjective probabilities, yet can be obtained by adjusting the full-investment solution with reference to $\mathbb{P}^Y$.

In what follows we analyze both regimes. The full-investment allocation provides the clean connection with the statistical formulation of betting and the classical (transaction-free) Kelly criterion, while the minimal-exposure allocation is included as an alternative replication motivated by transaction costs.

4.4. Agent's dichotomy

We have seen that the fund M corresponding to the agent's beliefs $\mathbb{P}^M$ is given by the likelihood ratio

$$M_Y(T, \omega) = \frac{p(\omega|M)}{p(\omega|Y)} M_Y(0).$$

However, the agent may choose not to replicate (or may be unable to replicate due to trading restrictions) this payoff, and instead replicates $V_Y(T, \omega)$. The corresponding state price density $\mathbb{P}^V$ of the portfolio V is

$$p(\omega|V) = \frac{V_Y(T, \omega)}{V_Y(0)} p(\omega|Y).$$

The central message of this paper, proved in the next section, is that the likelihood-ratio price $M_Y(T,\omega)$ maximizes logarithmic utility:

$$\mathbb{E}^M \log\left(\frac{M_Y(T)}{M_Y(0)}\right) = \max_{V_Y(T)} \mathbb{E}^M \log\left(\frac{V_Y(T)}{V_Y(0)}\right).$$

From this perspective, the portfolio M with state price density $\mathbb{P}^M$ is already log-utility optimal, leaving nothing further to optimize. This yields a straightforward solution for the log-utility optimal weights.

For general utility maximization,

$$\max_{V_Y(T)} \mathbb{E}^M\, U\left(\frac{V_Y(T)}{V_Y(0)}\right),$$

the optimal payoff $V_Y(T,\omega)$ has a state price density $\mathbb{P}^V$ different from $\mathbb{P}^M$ unless U is logarithmic. Statistically, utility maximization produces a new measure $\mathbb{P}^V$ as a blend of the physical $\mathbb{P}^M$ and risk-neutral $\mathbb{P}^Y$ measures. The resulting portfolio V is again log-utility optimal under its own state price density $\mathbb{P}^V$, allowing log-utility results to follow.

4.5. *Solution and alternative hedge representations*

Any payoff $V_Y(T,\omega)$ is achieved by setting

$$\Delta^i = V_Y(T,\omega_i),$$

in which case the portfolio at time T is

$$V_Y(T,\omega) = \sum_{i=1}^{N} V_Y(T,\omega_i)\, X_Y^i(T).$$

Since Arrow–Debreu securities are linearly dependent, the same payoff is achieved by holding Δ^Y units of a risk-free asset Y:

$$\begin{aligned} V_Y &= \sum_{i=1}^{N} \Delta^i X_Y^i \\ &= \sum_{i=1}^{N} \Delta^i X_Y^i - \Delta^Y \sum_{i=1}^{N} X_Y^i + \Delta^Y \\ &= \sum_{i=1}^{N} (\Delta^i - \Delta^Y)\, X_Y^i + \Delta^Y. \end{aligned}$$

Given linear dependence, the same payoff can be achieved with this ambiguity. If we further restrict hedging positions to be non-negative, the possible range for Δ^Y is $[0, \min_j \Delta^j]$. The exact choice makes no difference to the payoff in markets free of arbitrage and friction. The situation corresponding to the statistical approach is to be fully invested in the respective Arrow–Debreu securities, choosing $\Delta^Y = 0$. However, a traditional approach in the literature is to minimize positions in Arrow–Debreu securities while preserving non-negativity, effectively choosing $\Delta^Y_{\min} = \min_j \Delta^j$.

In the particular case of the fund M with the state price density corresponding to the beliefs $\mathbb{P}^M$, the same payoff $M_Y(T,\omega) = \frac{p(\omega|M)}{p(\omega|Y)} M_Y(0)$ is achieved either by trading fully in Arrow–Debreu securities X^i and keeping $\Delta^Y = 0$,

$$\begin{aligned} &(\Delta^1, \ldots, \Delta^N, \Delta^Y) \\ &\quad = \left(\frac{p(\omega_1|M)}{p(\omega_1|Y)} M_Y(0), \ldots, \frac{p(\omega_N|M)}{p(\omega_N|Y)} M_Y(0), 0\right), \end{aligned}$$

or by being maximally loaded in the risk-free asset, $\Delta^Y_{\min} = \min_j \frac{p(\omega_j|M)}{p(\omega_j|Y)} M_Y(0)$, leading to minimal positions in the $\Delta^i_{\min}$'s:

$$\begin{aligned} &(\Delta^1_{\min}, \ldots, \Delta^N_{\min}, \Delta^Y_{\min}) \\ &\quad = \left(\frac{p(\omega_1|M)}{p(\omega_1|Y)} M_Y(0) - \min_j \frac{p(\omega_j|M)}{p(\omega_j|Y)} M_Y(0),\right. \\ &\qquad \frac{p(\omega_2|M)}{p(\omega_2|Y)} M_Y(0) - \min_j \frac{p(\omega_j|M)}{p(\omega_j|Y)} M_Y(0), \ \ldots, \\ &\qquad \frac{p(\omega_N|M)}{p(\omega_N|Y)} M_Y(0) - \min_j \frac{p(\omega_j|M)}{p(\omega_j|Y)} M_Y(0), \\ &\qquad \left.\min_j \frac{p(\omega_j|M)}{p(\omega_j|Y)} M_Y(0)\right). \end{aligned}$$

In the case of full investment in Arrow–Debreu securities, $\Delta^i = \frac{p(\omega_i|M)}{p(\omega_i|Y)} M_Y(0)$, this representation becomes

$$\begin{aligned} M_Y(T,\omega_j) &= M_Y(0) + \sum_{i=1}^{N} \frac{p(\omega_i|M)}{p(\omega_i|Y)} \left(1_{\{\omega=\omega_j\}} - p(\omega_i|Y)\right) M_Y(0) \\ &= M_Y(0) + \frac{p(\omega_j|M)}{p(\omega_j|Y)} M_Y(0) - \sum_{i=1}^{N} p(\omega_i|Y)\, M_Y(0) \\ &= \frac{p(\omega_j|M)}{p(\omega_j|Y)} M_Y(0), \end{aligned}$$

confirming the link between the additive and multiplicative representations of the payoff.

In the alternative replication where the agent minimizes positions in Arrow–Debreu securities, the corresponding representation becomes

$$\begin{aligned} M_Y(T,\omega_j) &= M_Y(0) + \sum_{i=1}^{N} \left(\frac{p(\omega_i|M)}{p(\omega_i|Y)} - \min_k \frac{p(\omega_k|M)}{p(\omega_k|Y)}\right) \\ &\quad \times \left(1_{\{\omega=\omega_j\}} - p(\omega_i|Y)\right) M_Y(0) \\ &= M_Y(0) + \frac{p(\omega_j|M)}{p(\omega_j|Y)} M_Y(0) - \min_k \frac{p(\omega_k|M)}{p(\omega_k|Y)} M_Y(0) \\ &\quad - \sum_{i=1}^{N} p(\omega_i|Y)\, M_Y(0) + \sum_{i=1}^{N} p(\omega_i|Y) \min_k \frac{p(\omega_k|M)}{p(\omega_k|Y)} M_Y(0) \\ &= \frac{p(\omega_j|M)}{p(\omega_j|Y)} M_Y(0), \end{aligned}$$

so these two different hedging representations give the same payoff.

This approach is well illustrated by the classical Kelly criterion. The agent should get

$$M_Y(1,H) = \frac{p}{q} M_Y(0), \quad M_Y(1,T) = \frac{1-p}{1-q} M_Y(0),$$

so for the portfolio that trades fully in $\{X^H, X^T\}$ (i.e. $\Delta^Y = 0$),

$$\begin{aligned} M_Y(1,\omega) &= M_Y(1,H)\, X^H(\omega) + M_Y(1,T)\, X^T(\omega) \\ &= \frac{p}{q} M_Y(0)\, X^H(\omega) + \frac{1-p}{1-q} M_Y(0)\, X^T(\omega). \end{aligned}$$

Alternatively, the agent can reduce exposure in X^T by choosing $\Delta^Y = \frac{1-p}{1-q} M_Y(0)$, assuming $p > q$:

$$\begin{aligned} M_Y(1,\omega) &= \frac{p-q}{q(1-q)} M_Y(0)\, X^H(\omega) + \frac{1-p}{1-q} M_Y(0) \\ &= M_Y(0) + \frac{p-q}{q(1-q)} M_Y(0)\, \left(X^H(\omega) - q\right). \end{aligned}$$

Equivalently, the hedging positions can be decomposed in the underlying assets (X^H, X^T, Y). In one corner case (full investment in Arrow–Debreu securities), the positions are

$$(\Delta^H, \Delta^T, \Delta^Y) = \left(\frac{p}{q} M_Y(0), \frac{1-p}{1-q} M_Y(0), 0\right).$$

In the other corner case, with the position in X^T set to zero, the trading positions are

$$(\Delta^H_{\min}, \Delta^T_{\min}, \Delta^Y_{\min}) = \left(\frac{p-q}{q(1-q)} M_Y(0), 0, \frac{1-p}{1-q} M_Y(0)\right).$$

For instance, when $p = 0.6$, $q = 0.5$, $M_Y(0) = 100$, the payoff is

$$M_Y(1,\omega) = \begin{cases} \frac{p}{q} M_Y(0) = 120, & \omega = H, \\ \frac{1-p}{1-q} M_Y(0) = 80, & \omega = T. \end{cases}$$

This payoff is achieved equivalently by holding $(120, 80, 0)$ units of (X^H, X^T, Y), or by holding $(40, 0, 80)$ units of (X^H, X^T, Y).

In the soccer example with payoff given by equation (5), the case of full investment in Arrow–Debreu securities corresponds to

$$(\Delta^W, \Delta^D, \Delta^L, \Delta^Y) = (1200, 1000, 500, 0),$$

while the case of minimal positions in the Δ^i corresponds to

$$(\Delta^W_{\min}, \Delta^D_{\min}, \Delta^L_{\min}, \Delta^Y_{\min}) = (700, 500, 0, 500).$$

4.6. Relative positions

Another representation in relative terms uses w^i as the fraction of wealth invested in X^i:

$$\frac{M_Y(t)}{M_Y(0)} = \sum_{i=1}^{N} w^i \frac{X^i_Y(t)}{X^i(0)}, \quad w^i = \Delta^i \frac{X^i(0)}{M_Y(0)}.$$

In the case of full investment in Arrow–Debreu securities, $\Delta^i = \dfrac{p(\omega_i|M)}{p(\omega_i|Y)} M_Y(0)$ and $X^i(0) = \mathbb{E}^Y[X^i_Y(T)] = p(\omega_i|Y)$, so the corresponding weight invested in X^i simplifies to

$$w^i = \frac{p(\omega_i|M)}{p(\omega_i|Y)} M_Y(0) \cdot \frac{p(\omega_i|Y)}{M_Y(0)} = p(\omega_i|M).$$

While the absolute position $\Delta^i = \dfrac{p(\omega_i|M)}{p(\omega_i|Y)} M_Y(0)$ depends on market prices, the relative weight $w^i = p(\omega_i|M)$ is unilateral, mirroring the agent's subjective frequencies of market scenarios.

When the agent maximizes the position Δ^Y in the risk-free security Y, the corresponding relative weight becomes

$$\begin{aligned} w^i_{\min} &= \left(\frac{p(\omega_i|M)}{p(\omega_i|Y)} M_Y(0) - \min_j \frac{p(\omega_j|M)}{p(\omega_j|Y)} M_Y(0)\right) \frac{p(\omega_i|Y)}{M_Y(0)} \\ &= p(\omega_i|M) - p(\omega_i|Y) \cdot \min_j \frac{p(\omega_j|M)}{p(\omega_j|Y)}. \end{aligned} \tag{7}$$

For the classical Kelly criterion, the weights can be allocated as

$$(w^H, w^T, w^Y) = (p, 1-p, 0),$$

in the full-investment case, or equivalently as

$$(w^H_{\min}, w^T_{\min}, w^Y_{\min}) = \left(\frac{p-q}{1-q}, 0, \frac{1-p}{1-q}\right)$$

in the minimal-investment case.

For the soccer betting example, full investment corresponds to

$$(w^W, w^D, w^L, w^Y) = (p^W, p^D, p^L, 0) = (0.6, 0.3, 0.1, 0),$$

and the minimal-investment case to

$$\begin{aligned} &(w^W_{\min}, w^D_{\min}, w^L_{\min}, w^Y_{\min}) \\ &\quad = \left(p^W - q^W \cdot \frac{p^L}{q^L}, p^D - q^D \cdot \frac{p^L}{q^L}, p^L - q^L \cdot \frac{p^L}{q^L}, \frac{p^L}{q^L}\right) \\ &\quad = (0.35, 0.15, 0, 0.5) \end{aligned}$$

since $\frac{p^L}{q^L} = \min_j \frac{p(\omega_j|M)}{p(\omega_j|Y)}$.

Table 1 summarizes the allocation in Arrow–Debreu securities, both in absolute and relative terms, under the two corner cases: full investment and minimal investment in these securities.

Table 1. Comparison of allocations under full and minimal investment regimes.

	Full investment	Minimal investment
Δ^i	$\frac{p(\omega_i\|M)}{p(\omega_i\|Y)} M_Y(0)$	$\frac{p(\omega_i\|M)}{p(\omega_i\|Y)} M_Y(0) - \min_j \frac{p(\omega_j\|M)}{p(\omega_j\|Y)} M_Y(0)$
w^i	$p(\omega_i\|M)$	$p(\omega_i\|M) - p(\omega_i\|Y)\ \min_j \frac{p(\omega_j\|M)}{p(\omega_j\|Y)}$

5. Logarithmic utility optimality of the ratio

We now prove that the payoff

$$\frac{M_Y(T)}{M_Y(0)} = \frac{p(\omega|M)}{p(\omega|Y)}$$

is *log-utility optimal* under $\mathbb{P}^M$. This means it solves

$$\max_{V_Y(\cdot)} \mathbb{E}^M\left[\log\left(\frac{V_Y(T)}{V_Y(0)}\right)\right].$$

In other words, recognizing that $\frac{p(\omega|M)}{p(\omega|Y)}$ is a valid payoff immediately implies it is the growth-optimal portfolio (GOP). If an investor picks another utility function, such as isoelastic with risk-aversion γ, the resulting payoff yields a *different* implied measure $\mathbb{P}^V$—but only $\mathbb{P}^V$, not $\mathbb{P}^M$, is observed by the market.

A second reason the ratio $\frac{p(\omega|M)}{p(\omega|Y)}$ may not appear in practice is if it is *non-replicable* under real trading constraints. In that case, the investor seeks a nearest alternative $\mathbb{P}^V$ that *is* replicable, effectively minimizing $D_{\mathrm{KL}}(\mathbb{P}^M \parallel \mathbb{P}^V)$. This quantity—the relative entropy, also known as the Kullback–Leibler divergence (Kullback and Leibler 1951), is formalized in the next definition. This perspective unifies growth-optimal portfolios with Kullback–Leibler divergence as a measure of distance among distributions. We focus on the finite set of market scenarios Ω case; see Vecer *et al.* (2025) for continuous extensions.

Definition 1 Kullback–Leibler Divergence (Relative Entropy) *Let $\mathbb{P}$ and $\mathbb{Q}$ be probability measures on Ω. If $\mathbb{P}$ is absolutely continuous with respect to $\mathbb{Q}$ (denoted $\mathbb{P} \ll \mathbb{Q}$), the Kullback–Leibler divergence of $\mathbb{P}$ from $\mathbb{Q}$ is*

$$D_{\mathrm{KL}}(\mathbb{P} \parallel \mathbb{Q}) := \mathbb{E}^{\mathbb{P}}\left[\log\left(\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}}\right)\right].$$

In the finite-outcome setting $\Omega = \{\omega_1, \ldots, \omega_n\}$ with $\mathbb{P}(\omega_i) = p(\omega_i)$ and $\mathbb{Q}(\omega_i) = q(\omega_i)$, this becomes

$$D_{\mathrm{KL}}(\mathbb{P} \parallel \mathbb{Q}) = \sum_{i=1}^{n} p(\omega_i)\, \log\left(\frac{p(\omega_i)}{q(\omega_i)}\right),$$

with the convention $0\log(0/q) = 0$ and $D_{\mathrm{KL}}(\mathbb{P} \parallel \mathbb{Q}) = +\infty$ if $q(\omega) = 0$ for some ω with $p(\omega) > 0$.

Definition 2 Growth Optimal Portfolio *A portfolio V is called growth optimal with respect to a physical measure $\mathbb{P}^M$ if it maximizes*

$$\mathbb{E}^M\left[\log\left(\frac{V_Y(T)}{V_Y(0)}\right)\right].$$

5.1. *Properties of the growth optimal portfolio*

We first relate the expected logarithm of a general payoff V to the Kullback–Leibler divergence (relative entropy).

Theorem 5.1 *Let $\mathbb{P}^M$ be the physical measure, $\mathbb{P}^Y$ the risk-neutral measure, and V any portfolio with implied measure $\mathbb{P}^V$. Then*

$$\mathbb{E}^M\left[\log\left(\frac{V_Y(T)}{V_Y(0)}\right)\right] = D_{\mathrm{KL}}\left(\mathbb{P}^M \parallel \mathbb{P}^Y\right) - D_{\mathrm{KL}}\left(\mathbb{P}^M \parallel \mathbb{P}^V\right). \tag{8}$$

Proof Observe

$$\begin{aligned}
\mathbb{E}^M\left[\log\left(\frac{V_Y(T)}{V_Y(0)}\right)\right] &= \mathbb{E}^M\left[\log\left(\frac{p(\omega|V)}{p(\omega|Y)}\right)\right] \\
&= \int_\Omega p(\omega|M)\log\left(\frac{p(\omega|V)}{p(\omega|Y)}\right)\mathrm{d}\omega \\
&= \int_\Omega p(\omega|M)\log\left(\frac{p(\omega|M)}{p(\omega|Y)}\right)\mathrm{d}\omega \\
&\quad - \int_\Omega p(\omega|M)\log\left(\frac{p(\omega|M)}{p(\omega|V)}\right)\mathrm{d}\omega \\
&= D_{\mathrm{KL}}\left(\mathbb{P}^M \parallel \mathbb{P}^Y\right) - D_{\mathrm{KL}}\left(\mathbb{P}^M \parallel \mathbb{P}^V\right).
\end{aligned}$$

■

5.1.1. Immediate optimality.

Corollary 5.2 *The expected logarithmic return of the 'Kelly payoff' $\frac{M_Y(T)}{M_Y(0)}$ under $\mathbb{P}^M$ is precisely the relative entropy*

$$\mathbb{E}^M\left[\log\left(\frac{M_Y(T)}{M_Y(0)}\right)\right] = D_{\mathrm{KL}}\left(\mathbb{P}^M \parallel \mathbb{P}^Y\right), \tag{9}$$

and $\frac{M_Y(T)}{M_Y(0)}$ maximizes $\mathbb{E}^M\left[\log\left(\frac{V_Y(T)}{V_Y(0)}\right)\right]$.

Proof From Theorem 5.1, we have

$$\mathbb{E}^M\left[\log\left(\frac{M_Y(T)}{M_Y(0)}\right)\right] = D_{\mathrm{KL}}\left(\mathbb{P}^M \parallel \mathbb{P}^Y\right),$$

and for any other V,

$$\begin{aligned}
\mathbb{E}^M\left[\log\left(\frac{V_Y(T)}{V_Y(0)}\right)\right] &= D_{\mathrm{KL}}\left(\mathbb{P}^M \parallel \mathbb{P}^Y\right) - D_{\mathrm{KL}}\left(\mathbb{P}^M \parallel \mathbb{P}^V\right) \\
&\le D_{\mathrm{KL}}\left(\mathbb{P}^M \parallel \mathbb{P}^Y\right).
\end{aligned}$$

Thus $\frac{M_Y(T)}{M_Y(0)}$ attains the maximum. ■

In our numerical example with $\mathbb{P}^Y = (0.5, 0.3, 0.2)$ and $\mathbb{P}^M = (0.6, 0.3, 0.1)$, the portfolio achieving the highest growth rate corresponds to the likelihood ratio of the two measures $\mathbb{P}^M$ and $\mathbb{P}^Y$. Specifically, with $M(0) = 1000$, the payoff

$$\begin{aligned}
&(M_Y(1,\mathrm{Win}), M_Y(1,\mathrm{Draw}), M_Y(1,\mathrm{Loss})) \\
&= (1200, 1000, 500),
\end{aligned}$$

attains the maximal growth rate among all portfolios:

$$\begin{aligned}
&\mathbb{E}^M\left[\log\left(\frac{M_Y(1)}{M_Y(0)}\right)\right] \\
&= \mathbb{E}^M\left[\log\left(\frac{\mathbb{P}^M}{\mathbb{P}^Y}\right)\right] \\
&= \log\left(\frac{0.6}{0.5}\right)\cdot 0.6 + \log\left(\frac{0.3}{0.3}\right)\cdot 0.3 + \log\left(\frac{0.1}{0.2}\right)\cdot 0.1 \\
&\approx 0.1823 \times 0.6 + 0 \times 0.3 - 0.6931 \times 0.1 \\
&\approx 0.0401.
\end{aligned}$$

Consider instead a modified payoff

$$V_Y(1) = (1200, 1000, 800),$$

which requires initial capital $V_Y(0) = 1060$. By equation (8), its growth rate is

$$\mathbb{E}^M \log\left(\frac{V_Y(1)}{V_Y(0)}\right) = D_{KL}\left(\mathbb{P}^M \parallel \mathbb{P}^Y\right) - D_{KL}\left(\mathbb{P}^M \parallel \mathbb{P}^V\right),$$

diminished by the relative entropy of $\mathbb{P}^M$ with respect to $\mathbb{P}^V$. Concretely,

$$\begin{aligned}
&D_{KL}\left(\mathbb{P}^M \parallel \mathbb{P}^V\right)\\
&\quad = \mathbb{E}^M\left[\log\left(\frac{\mathbb{P}^M}{\mathbb{P}^V}\right)\right] = \sum_{\omega} \mathbb{P}^M(\omega)\,\log\left(\frac{\mathbb{P}^M(\omega)}{\mathbb{P}^V(\omega)}\right)\\
&\quad \approx 0.6\,\log\left(\frac{0.6}{0.56604}\right) + 0.3\,\log\left(\frac{0.3}{0.28302}\right)\\
&\qquad + 0.1\,\log\left(\frac{0.1}{0.15094}\right)\\
&\quad \approx 0.6 \times 0.0583 + 0.3 \times 0.0583 + 0.1 \times (-0.4117)\\
&\quad \approx 0.0113.
\end{aligned}$$

Hence, the portfolio $V(1)$ has an expected growth rate of

$$\mathbb{E}^M \log\left(\frac{V_Y(1)}{V_Y(0)}\right) = 0.0401 - 0.0113 = 0.0288,$$

or 2.88%. This illustrates how reducing the drawdown on the 'Loss' scenario leads to a lower growth rate than that of the growth-optimal portfolio M.

We focus on the setting of *prediction markets* where investing $p(\omega|M)$ fraction of wealth in each Arrow–Debreu security perfectly replicates the desired payoff. Since these positions are nonnegative, there is no need for shorting. In general financial markets, however, not all Arrow–Debreu contracts may be traded; the investor may have to choose from among available assets, each with state-price density $\mathbb{P}^X$. In that case, the agent can only form mixtures

$$p(\omega|V) = \sum_{i=1}^{N} w^i\, p(\omega|X^i),$$

where w^i are the fractions of wealth invested in each asset X^i. Perfect replication of $\mathbb{P}^M$ might not be feasible, so the agent settles for the best $\mathbb{P}^V$.

Since $D_{\mathrm{KL}}\big(\mathbb{P}^M \parallel \mathbb{P}^Y\big)$ is constant, maximizing $\mathbb{E}^M[\log(\cdot)]$ reduces to *minimizing* $D_{\mathrm{KL}}(\mathbb{P}^M \parallel \mathbb{P}^V)$. Thus the growth-optimal portfolio is the 'closest' replicable state price density to $\mathbb{P}^M$ in relative entropy:

THEOREM 5.3 *If V is growth optimal under $\mathbb{P}^M$, then its implied measure $\mathbb{P}^V$ satisfies*

$$D_{\mathrm{KL}}\big(\mathbb{P}^M \parallel \mathbb{P}^V\big) = \min_{\mathbb{P}^P \text{ replicable}} D_{\mathrm{KL}}\big(\mathbb{P}^M \parallel \mathbb{P}^P\big).$$

Proof Follows directly from equation (8). ■

COROLLARY 5.4 Numéraire Invariance *The growth-optimal payoff is invariant under the choice of benchmark asset. That is, for any reference asset X,*

$$\max_{V_X(\cdot)} \mathbb{E}^M\left[\log\left(\frac{V_X(T)}{V_X(0)}\right)\right]$$

is achieved by the same payoff (up to a change of units).

COROLLARY 5.5 *For any asset X, the growth-optimal portfolio under $\mathbb{P}^X$ is X itself (i.e. invests fully in X).*

Proof Choosing $\mathbb{P}^V = \mathbb{P}^X$ drives $D_{\mathrm{KL}}(\mathbb{P}^X \parallel \mathbb{P}^V) = 0$. ■

In particular, every portfolio V is growth-optimal with respect to its own state price density $\mathbb{P}^V$. For example, consider a portfolio V with payoff $V_Y(1) = (1200, 1000, 800)$ and initial price $V_Y(0) = 1060$. This portfolio achieves its maximal growth rate relative to $\mathbb{P}^V = (0.56604, 0.28302, 0.15094)$:

$$\begin{aligned}
&\mathbb{E}^V\left[\log\left(\frac{V_Y(1)}{V_Y(0)}\right)\right]\\
&\quad = \mathbb{E}^V\left[\log\left(\frac{\mathbb{P}^V}{\mathbb{P}^Y}\right)\right]\\
&\quad = \log\left(\frac{0.56604}{0.5}\right)\cdot 0.56604 + \log\left(\frac{0.28302}{0.3}\right)\cdot 0.28302\\
&\qquad + \log\left(\frac{0.15094}{0.2}\right)\cdot 0.15094\\
&\quad \approx 0.1241 \times 0.56604 - 0.0583 \times 0.28302\\
&\qquad - 0.2841 \times 0.15094\\
&\quad \approx 0.0113.
\end{aligned}$$

By contrast, portfolio M has a reduced growth return under $\mathbb{P}^V$ due to the relative entropy $D_{KL}(\mathbb{P}^V \parallel \mathbb{P}^M)$. Specifically,

$$\begin{aligned}
\mathbb{E}^V\left[\log\left(\frac{M_Y(1)}{M_Y(0)}\right)\right] &= D_{KL}\big(\mathbb{P}^V \parallel \mathbb{P}^Y\big) - D_{KL}\big(\mathbb{P}^V \parallel \mathbb{P}^M\big)\\
&\approx 0.0113 - 0.0127 = -0.0014.
\end{aligned}$$

From a statistical perspective on likelihood modeling, any candidate distribution $\mathbb{P}^M$ is valid. From the viewpoint of the physical measure $\mathbb{P}^M$, the ratio $\frac{\mathbb{P}^M}{\mathbb{P}^Y}$ represents the scaled growth-optimal portfolio specified by the payoff $\frac{M_Y(1)}{M_Y(0)}$. An agent can specify a payoff $V_Y(1)$, determining its implied state price density $\mathbb{P}^V$, under which V is growth-optimal. Hence, the agent always produces a growth-optimal portfolio. Because all portfolios are inherently growth-optimal under their own state price densities, the real question is not how to optimize the payoff, but rather which statistical measure $\mathbb{P}^M$ one should choose initially.

5.2. *Utility maximization and tempered measures*

A standard approach is to solve the optimization problem

$$\max_V \; \mathbb{E}^M[U(V_Y(T))].$$

For a strictly concave utility function U on $(0,\infty)$, classical results (e.g. Kramkov and Schachermayer 1999, Karatzas and Shreve 1998) show that the optimal payoff can be written as

$$V_Y(T,\omega) = I\left(\lambda \frac{p(\omega|Y)}{p(\omega|M)}\right), \quad I(x) = (U')^{-1}(x), \tag{10}$$

where $\lambda > 0$ is chosen to meet the budget constraint $\mathbb{E}^Y[V_Y(T)] = V_Y(0)$. By equation (4), this induces an implied measure $\mathbb{P}^V$:

$$p(\omega|V) = \frac{I\left(\lambda \frac{p(\omega|Y)}{p(\omega|M)}\right) p(\omega|Y)}{\sum_{\omega'\in\Omega} I\left(\lambda \frac{p(\omega'|Y)}{p(\omega'|M)}\right) p(\omega'|Y)}. \tag{11}$$

Hence, the payoff can be viewed as a 'tempering' of the pure Kelly ratio $\frac{p(\omega|M)}{p(\omega|Y)}$.

An especially transparent instance is *isoelastic* (power) utility, also known as the CRRA (constant relative risk aversion) class,

$$U(x) = \frac{x^{1-\gamma}-1}{1-\gamma}, \quad \gamma > 0,$$

where the coefficient γ represents the agent's relative risk aversion. In this case, the implied measure is

$$p(\omega|V) = \frac{\left(\frac{p(\omega|M)}{p(\omega|Y)}\right)^{\frac{1}{\gamma}} p(\omega|Y)}{\sum_{\omega'\in\Omega}\left(\frac{p(\omega'|M)}{p(\omega'|Y)}\right)^{\frac{1}{\gamma}} p(\omega'|Y)}. \tag{12}$$

As $\gamma \to \infty$, $\mathbb{P}^V$ converges to $\mathbb{P}^Y$ (infinite risk aversion). For small γ, $\mathbb{P}^V$ remains closer to $\mathbb{P}^M$. In particular, $\gamma = 1$ corresponds to logarithmic utility, yielding $\mathbb{P}^V = \mathbb{P}^M$. This independently confirms the result from Corollary 5.2, namely that the payoff given by the likelihood ratio $\frac{\mathbb{P}^M}{\mathbb{P}^Y}$ is logarithmically optimal.

5.2.1. Identifying minimal weights. The probabilities in equation (12) correspond to the *full-investment weights*, i.e. the fractions of wealth allocated directly to Arrow–Debreu securities. When instead we adopt the *minimal-exposure* representation (cf. section 4), the weights are adjusted as

$$w^i_{\min} = p(\omega_i|V) - p(\omega_i|Y)\cdot \min_j \frac{p(\omega_j|V)}{p(\omega_j|Y)}. \tag{13}$$

This representation preserves the same payoff but reduces exposure in Arrow–Debreu securities by maximizing the position in the risk-free asset. While the formula is clean, it has been overlooked in the literature because its key ingredient, $p(\omega_i|V)$ from equation (12), already involves a highly nonlinear transformation of the probabilities.

The preceding discussion yields explicit allocation formulas under CRRA preferences. For clarity, we summarize the result in the following proposition, which also covers the logarithmic case as $\gamma = 1$.

PROPOSITION 5.6 CRRA Allocations in Arrow–Debreu Securities *Consider an agent with CRRA utility*

$$U(x) = \frac{x^{1-\gamma}-1}{1-\gamma}, \quad \gamma > 0,$$

subjective probabilities $\mathbb{P}^M = (p(\omega|M))_{\omega\in\Omega}$, *and market probabilities* $\mathbb{P}^Y = (p(\omega|Y))_{\omega\in\Omega}$. *The optimal terminal payoff induces the implied measure*

$$p(\omega|V) = \frac{\left(\frac{p(\omega|M)}{p(\omega|Y)}\right)^{1/\gamma} p(\omega|Y)}{\sum_{\omega'}\left(\frac{p(\omega'|M)}{p(\omega'|Y)}\right)^{1/\gamma} p(\omega'|Y)}.$$

This leads to two equivalent hedge representations: Absolute positions:

$$\Delta^i = \frac{p(\omega_i|V)}{p(\omega_i|Y)} V_Y(0), \quad \Delta^i_{\min} = \Delta^i - \min_j \Delta^j,$$

with the residual invested in the risk-free asset Y. Relative weights:

$$w^i = p(\omega_i|V), \quad w^i_{\min} = p(\omega_i|V) - p(\omega_i|Y)\cdot \min_j \frac{p(\omega_j|V)}{p(\omega_j|Y)}.$$

REMARK 1 Log Utility as Special Case When $\gamma = 1$, we have $p(\omega|V) = p(\omega|M)$, so the allocations reduce to the classical Kelly criterion based on the likelihood ratio. In particular:

(i) *Full investment:*

$$\Delta^i = \frac{p(\omega_i|M)}{p(\omega_i|Y)} M_Y(0), \quad w^i = p(\omega_i|M).$$

(ii) *Minimal exposure:*

$$\Delta^i_{\min} = \frac{p(\omega_i|M)}{p(\omega_i|Y)} M_Y(0) - \min_j \frac{p(\omega_j|M)}{p(\omega_j|Y)} M_Y(0),$$

$$w^i_{\min} = p(\omega_i|M) - p(\omega_i|Y)\cdot \min_j \frac{p(\omega_j|M)}{p(\omega_j|Y)}.$$

Thus, the log-utility case (Kelly) is embedded as a special case of the CRRA formulation, with both absolute positions Δ^i and relative weights w^i obtained in closed form.

5.2.2. Soccer example. Consider the soccer example with $\mathbb{P}^M = (0.6, 0.3, 0.1)$ and $\mathbb{P}^Y = (0.5, 0.3, 0.2)$. For $\gamma = 2$, the implied measure from equation (12) is

$$\mathbb{P}^V = (0.5537, 0.3033, 0.1430),$$

which corresponds to the full-investment allocation. Applying equation (13), the minimal weights are

$$w_{\min} = (0.1963, 0.0888, 0),$$

illustrating how the representation shifts part of the exposure into the risk-free asset while preserving the same payoff structure.

5.2.3. Binary example and fractional Kelly. In the binary setting $\Omega = \{H, T\}$ with $\mathbb{P}^M = (p, 1-p)$ and $\mathbb{P}^Y = (q, 1-q)$, the isoelastic optimum is

$$p(\omega|V) = \begin{cases} \dfrac{\left(\frac{p}{q}\right)^{\frac{1}{\gamma}} q}{\left(\frac{p}{q}\right)^{\frac{1}{\gamma}} q + \left(\frac{1-p}{1-q}\right)^{\frac{1}{\gamma}} (1-q)} \approx q + \frac{p-q}{\gamma}, & \omega = H, \\ \dfrac{\left(\frac{1-p}{1-q}\right)^{\frac{1}{\gamma}} (1-q)}{\left(\frac{p}{q}\right)^{\frac{1}{\gamma}} q + \left(\frac{1-p}{1-q}\right)^{\frac{1}{\gamma}} (1-q)} \approx (1-q) - \frac{p-q}{\gamma}, & \omega = T. \end{cases} \tag{14}$$

When $\gamma = 1$, $\mathbb{P}^V = \mathbb{P}^M$, the pure Kelly solution; for large γ, $\mathbb{P}^V$ converges to $\mathbb{P}^Y$. From this perspective, the fractional-Kelly bet (Davis and Lleo 2013) – investing only $\frac{1}{\gamma}$ times the full Kelly fraction $\frac{p-q}{1-q}$—is a first-order approximation to equation (14).

In all these cases, the market sees only $\mathbb{P}^V$. One could simply *start* with that as a final 'model' without referencing $\mathbb{P}^M$. Nevertheless, from a purely statistical point of view, altering $\mathbb{P}^M$ to $\mathbb{P}^V$ is ad hoc unless justified by external preferences or constraints.

6. Stock market

In contrast to prediction markets, which trade each market outcome individually, one can conceptualize that each asset in the market X is represented by a full probabilistic measure $\mathbb{P}^X$, known as the *state price density*. By purchasing the asset X, the agent effectively acquires the entire portfolio of all individual outcomes ω, each weighted by $\mathbb{P}^X(\omega)$. Because these fractions are implicitly specified for all outcomes jointly, the agent may be unable to perfectly match her physical measure $\mathbb{P}^M$ by simply constructing portfolios of individual assets.

Using the same arguments that lead to equation (3), the price evolution of the relative price $X_Y(T, \omega)$ of two assets X and Y can be represented as a scaled ratio of their state price densities $\mathbb{P}^X$ and $\mathbb{P}^Y$:

$$X_Y(T, \omega) = \frac{p(\omega|X)}{p(\omega|Y)} \cdot X_Y(0).$$

This approach is discussed in more detail in Vecer *et al.* (2025). In particular, the binary model $\mathbb{P}^X = (p, 1-p)$ and $\mathbb{P}^Y = (q, 1-q)$ recovers the binomial model

$$X_Y(1, \omega) = \begin{cases} \frac{p}{q} \cdot X_Y(0) = u \cdot X_Y(0), & \omega = H, \\ \frac{1-p}{1-q} \cdot X_Y(0) = d \cdot X_Y(0), & \omega = T, \end{cases}$$

where $u = \frac{p}{q}$ and $d = \frac{1-p}{1-q}$.

The normal model for the simple return process

$$dR(t) = \frac{dX_Y(t)}{X_Y(t)}$$

assumes $R(T) \sim N(0, \sigma^2 T)$ under the risk-neutral distribution $\mathbb{P}^Y$ and $R(T) \sim N(\sigma^2 T, \sigma^2 T)$ under the stock measure $\mathbb{P}^X$, yielding the geometric Brownian motion model

$$\begin{aligned} X_Y(T, R(T)) &= \frac{p\,(R(T)|X)}{p\,(R(T)|Y)} \cdot X_Y(0) \\ &= \frac{\frac{1}{\sqrt{2\sigma^2 T}} \exp\left(-\frac{(R(T)-\sigma^2 T)^2}{2\sigma^2 T}\right)}{\frac{1}{\sqrt{2\sigma^2 T}} \exp\left(-\frac{R(T)^2}{2\sigma^2 T}\right)} \cdot X_Y(0) \\ &= \exp\left(R(T) - \frac{1}{2}\sigma^2 T\right) \cdot X_Y(0) \\ &= \exp\left(\sigma\, W(T) - \frac{1}{2}\sigma^2 T\right) \cdot X_Y(0). \end{aligned}$$

The agent can create portfolios V as linear combinations of positions in the individual assets. Considering two assets, the static portfolios have the form

$$\frac{V_Y(T, \omega)}{V_Y(0)} = w^X \cdot \frac{X_Y(T, \omega)}{X_Y(0)} + w^Y \cdot \frac{Y_Y(T, \omega)}{Y_Y(0)},$$

where w^X and w^Y are the fractions of wealth invested in the assets X and Y, with the condition $w^X + w^Y = 1$. Trivially, the state price density of the portfolio V is the mixture distribution

$$\mathbb{P}^V = w^X \cdot \mathbb{P}^X + w^Y \cdot \mathbb{P}^Y.$$

According to Theorem 5.3, the goal of finding the growth optimal portfolio is equivalent to minimizing

$$\begin{aligned} &\min_{\mathbb{P}^V} D_{KL}\left(\mathbb{P}^M \,\|\, \mathbb{P}^V\right) \\ &\quad = \min_{w^X} D_{KL}\left(\mathbb{P}^M \,\|\, w^X \cdot \mathbb{P}^X + (1 - w^X) \cdot \mathbb{P}^Y\right), \end{aligned}$$

meaning that we seek the mixture distribution $\mathbb{P}^V$ that is closest, in the Kullback–Leibler sense, to the agent's physical measure $\mathbb{P}^M$.

Because the distributions $\mathbb{P}^X$ and $\mathbb{P}^Y$ act like full fixed lists of prices for given market scenarios, one may no longer be able to replicate arbitrary distributional views $\mathbb{P}^M$ perfectly by using only linear mixtures of the state price densities.

Let us provide the minimal example showing when a perfect match is not possible due to market incompleteness. Consider a trinomial model, where the state price density $\mathbb{P}^X = (p_1^X, p_2^X, p_3^X)$ and the state price density $\mathbb{P}^Y = (p_1^Y, p_2^Y, p_3^Y)$. This corresponds to the trinomial price evolution

$$X_Y(1, \omega) = \begin{cases} \dfrac{p_1^X}{p_1^Y} \cdot X_Y(0), & \omega = \omega_1, \\ \dfrac{p_2^X}{p_2^Y} \cdot X_Y(0), & \omega = \omega_2, \\ \dfrac{p_3^X}{p_3^Y} \cdot X_Y(0), & \omega = \omega_3. \end{cases}$$

The agent with the physical measure $\mathbb{P}^M = (p_1^M, p_2^M, p_3^M)$ seeks to minimize

$$\min_{w^X} \sum_{i=1}^{3} \log\left(\frac{p_i^M}{w^X\, p_i^X + (1 - w^X)\, p_i^Y}\right) p_i^M. \tag{15}$$

Due to the convexity of relative entropy, the solution w^X is unique. Taking the derivative with respect to w^X, the optimal solution satisfies

$$\sum_{i=1}^{3} \frac{p_i^X - p_i^Y}{w^X p_i^X + (1-w^X) p_i^Y} p_i^M = 0,$$

which admits a closed-form solution by solving a quadratic equation, although the general form can be quite involved even in this simple example.

Under the simplifying assumption $p_2^X = p_2^Y$, the formula for the optimal w^X becomes

$$w^X = -\frac{p_1^M (p_1^X - p_1^Y) p_3^Y + p_3^M (p_3^X - p_3^Y) p_1^Y}{p_1^M (p_1^X - p_1^Y) \left(p_3^X - p_3^Y\right) + p_3^M (p_3^X - p_3^Y) \left(p_1^X - p_1^Y\right)}.$$

As an example, consider $\mathbb{P}^X = \left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right)$, $\mathbb{P}^Y = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$ and $\mathbb{P}^M = \left(\frac{1}{2}, \frac{1}{6}, \frac{1}{3}\right)$. This corresponds to the trinomial price evolution

$$X_Y(1,\omega) = \begin{cases} \frac{3}{2} \cdot X_Y(0), & \omega = \omega_1, \\ X_Y(0), & \omega = \omega_2, \\ \frac{1}{2} \cdot X_Y(0), & \omega = \omega_3, \end{cases}$$

with an uptick $u = \frac{3}{2}$ for scenario ω_1, a downtick $d = \frac{1}{2}$ for scenario ω_3, and a neutral scenario ω_2. The optimal weight (i.e. the fraction of wealth invested in the asset X) is $w^X = 0.4$, giving

$$\mathbb{P}^V = \frac{2}{5} \cdot \mathbb{P}^X + \frac{3}{5} \cdot \mathbb{P}^Y = \left(\frac{6}{15}, \frac{5}{15}, \frac{4}{15}\right).$$

The resulting optimal portfolio is

$$V_Y(1,\omega) = \frac{p(\omega|V)}{p(\omega|Y)} \cdot V_Y(0) = \begin{cases} \frac{6}{5} \cdot V_Y(0), & \omega = \omega_1, \\ V_Y(0), & \omega = \omega_2, \\ \frac{4}{5} \cdot V_Y(0), & \omega = \omega_3. \end{cases}$$

This portfolio V maximizes the expected logarithmic return $\mathbb{E}^M\left[\log\left(\frac{V_Y(1)}{V_Y(0)}\right)\right]$. Specifically, from equation (8), we obtain

$$\begin{aligned}\mathbb{E}^M\left[\log\left(\frac{V_Y(T)}{V_Y(0)}\right)\right] &= D_{\mathrm{KL}}\left(\mathbb{P}^M \parallel \mathbb{P}^Y\right) - D_{\mathrm{KL}}\left(\mathbb{P}^M \parallel \mathbb{P}^V\right) \\ &= 0.0872 - 0.0704 = 0.0168.\end{aligned}$$

Hence, were the agent permitted to invest in individual outcomes, she could realize $D_{\mathrm{KL}}\left(\mathbb{P}^M \parallel \mathbb{P}^Y\right) = 0.0872$. However, because the market restricts her to trade only linear combinations of the assets X and Y (i.e. mixtures of the distributions $\mathbb{P}^X$ and $\mathbb{P}^Y$), the closest replicable proxy is $\mathbb{P}^V$, which is relatively distant in terms of relative entropy, $D_{\mathrm{KL}}\left(\mathbb{P}^M \parallel \mathbb{P}^V\right) = 0.0704$. This acts as a penalizing factor on the expected log return of the entire portfolio, leaving her with an expected log return of 0.0168, or about 1.68%.

7. Link to mean-variance optimization

In practice, directly minimizing the relative entropy $D_{\mathrm{KL}}\left(\mathbb{P}^M \parallel \mathbb{P}^V\right)$ over all replicable state-price densities $\mathbb{P}^V$ (see equation (8) and the related discussion) can become analytically intractable. A useful alternative is to apply a second-order (Taylor) expansion of the logarithm, which leads naturally to a mean–variance form.

Suppose there are N tradable assets $X^1, \ldots, X^N$, each with an implied measure $\mathbb{P}^i$. Any static portfolio V formed at $t = 0$ by investing weights $\mathbf{w} = (w^1, \ldots, w^N)$ in these N assets satisfies

$$\mathbb{P}^V = \sum_{i=1}^{N} w^i \, \mathbb{P}^i, \qquad \frac{V_Y(T)}{V_Y(0)} = \sum_{i=1}^{N} w^i \frac{X_Y^i(T)}{X_Y^i(0)}.$$

Define the (under $\mathbb{P}^M$) *expected simple return* vector

$$\boldsymbol{\mu} = \left(\mathbb{E}^M\left[\frac{X_Y^1(T)}{X_Y^1(0)} - 1\right], \ldots, \mathbb{E}^M\left[\frac{X_Y^N(T)}{X_Y^N(0)} - 1\right]\right),$$

and the (under $\mathbb{P}^Y$) *covariance matrix*

$$\boldsymbol{\Sigma} = \left(\mathbb{E}^Y\left[\left(\frac{X_Y^i(T)}{X_Y^i(0)} - 1\right)\left(\frac{X_Y^j(T)}{X_Y^j(0)} - 1\right)\right]\right)_{1 \le i,j \le N}.$$

This approach has been introduced by Markowitz (1952).

For a portfolio V with terminal payoff $V_Y(T)$ and initial cost $V_Y(0)$, consider the expectation of its log-return under $\mathbb{P}^M$. Splitting it into parts under $\mathbb{P}^M$ and $\mathbb{P}^Y$, then applying a second-order expansion $\log(x) \approx (x-1) - \frac{1}{2}(x-1)^2$, gives:

$$\begin{aligned}\mathbb{E}^M\left[\log\left(\frac{V_Y(T)}{V_Y(0)}\right)\right] &= \left(\mathbb{E}^M - \mathbb{E}^Y\right)\left[\log\left(\frac{V_Y(T)}{V_Y(0)}\right)\right] \\ &\quad + \mathbb{E}^Y\left[\log\left(\frac{V_Y(T)}{V_Y(0)}\right)\right] \\ &\approx \left(\mathbb{E}^M - \mathbb{E}^Y\right)\left[\frac{V_Y(T)}{V_Y(0)} - 1\right] \\ &\quad - \tfrac{1}{2}\mathbb{E}^Y\left[\left(\frac{V_Y(T)}{V_Y(0)} - 1\right)^2\right] \\ &= \mathbf{w}^\top \boldsymbol{\mu} - \tfrac{1}{2}\mathbf{w}^\top \boldsymbol{\Sigma}\,\mathbf{w}. \end{aligned} \tag{16}$$

Maximizing this quadratic over all feasible $\mathbf{w}$ yields an *approximately* growth-optimal portfolio.

Taking the gradient with respect to $\mathbf{w}$ in $\mathbf{w}^\top \boldsymbol{\mu} - \frac{1}{2}\mathbf{w}^\top \boldsymbol{\Sigma}\,\mathbf{w}$, the unconstrained solution satisfies

$$\boldsymbol{\Sigma}\,\mathbf{w} = \boldsymbol{\mu}, \quad \text{so} \quad \mathbf{w} = \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu},$$

provided $\boldsymbol{\Sigma}$ is invertible on the desired subspace. In certain cases $\boldsymbol{\Sigma}$ is singular—for instance, if it includes a riskless asset or considering all Arrow–Debreu payoffs that are linearly dependent. One then solves $\boldsymbol{\Sigma}\,\mathbf{w} = \boldsymbol{\mu}$ in that affine subspace, using the *kernel* of $\boldsymbol{\Sigma}$ to impose $\sum_i w^i = 1$.

7.1. Example: Arrow–Debreu securities

Each Arrow–Debreu security X^i pays 1 only if scenario ω_i occurs (and 0 otherwise). Its price is $p(\omega_i|Y)$, so the expected simple return under $\mathbb{P}^M$ is

$$\mu_i = \mathbb{E}^M\left[\frac{1}{p(\omega_i|Y)} - 1\right] = \frac{p(\omega_i|M)}{p(\omega_i|Y)} - 1.$$

Because two distinct Arrow–Debreu securities cannot both pay 1 in the same scenario,

$$\begin{aligned}\mathbf{\Sigma}_{ij} &= \mathbb{E}^Y\left[\left(\frac{X_Y^i(T)}{X_Y^i(0)} - 1\right)\left(\frac{X_Y^j(T)}{X_Y^j(0)} - 1\right)\right] \\ &= \begin{cases}\dfrac{1}{p(\omega_i|Y)} - 1, & \text{if } i = j, \\ -1, & \text{if } i \neq j.\end{cases}\end{aligned}$$

The measure $\mathbb{P}^Y$ lies in the kernel of $\mathbf{\Sigma}$. A particular solution of $\mathbf{\Sigma}\,\mathbf{w} = \boldsymbol{\mu}$ that also satisfies $\sum_i w^i = 1$ is precisely $\mathbf{w} = (p(\omega_1|M), \ldots, p(\omega_n|M))$. Thus, even from a mean–variance perspective, buying each outcome in proportion to $p(\omega|M)$ is the optimal strategy. Notably, this means taking a positive fraction $p(\omega_i|M)$ in a scenario where $\frac{p(\omega_i|M)}{p(\omega_i|Y)} - 1 < 0$, because diversification against other scenarios reduces variance enough to offset that negative expected simple return.

Hence, the exact ratio $\frac{\mathbb{P}^M(\omega)}{\mathbb{P}^Y(\omega)}$ seen in section 3 is recovered here at first order by a Taylor approximation to the log objective. In more general incomplete markets, one can only approximate this ratio over the feasible span of assets, and the mean–variance approach from equation (16) offers a practical method to identify a close-to-optimal combination.

8. Likelihood view on good and bad properties

A likelihood-based interpretation of the Kelly criterion provides both simplicity and conceptual clarity. In this framework, an agent specifies a probability measure $\mathbb{P}^V$ that governs her wealth allocation across possible market scenarios. If the market trades individual outcomes, the agent allocates the fraction $\mathbb{P}^V(\omega)$ of her wealth to each outcome ω. If the market instead trades assets with fixed state price densities, she constructs a portfolio whose state price density is a convex combination of the component densities, consistent with the probabilities in $\mathbb{P}^V$.

Although the agent may hold a distinct subjective measure $\mathbb{P}^M$, only $\mathbb{P}^V$ is revealed to the market via the resulting payoff. In this view, portfolio optimization reduces to choosing the measure $\mathbb{P}^V$. Once $\mathbb{P}^V$ is set, the corresponding payoff is automatically the log-optimal (growth-optimal) portfolio under $\mathbb{P}^V$, so no further optimization is required.

A primary concern, as discussed by MacLean *et al.* (2011), is the possibility of large losses – often termed 'black swan' events—for which the agent's subjective belief $\mathbb{P}^M$ assigns a substantially lower probability than the market measure $\mathbb{P}^Y$. Formally, we define:

Definition 3 *A black swan outcome of magnitude order N is any outcome ω for which*

$$\log \mathbb{P}^M(\omega) - \log \mathbb{P}^Y(\omega) < -N.$$

In such scenarios, the agent's final wealth is at most e^{-N} of its initial value (equivalently, the drawdown is at least $1 - e^{-N}$). To reduce exposure to these extreme events, one can adopt a new communicated measure $\mathbb{P}^V$ defined as a convex combination of the physical and risk-neutral measures $\mathbb{P}^M$ and $\mathbb{P}^Y$. A natural approach, motivated by the approximation of the state price density that maximizes isoelastic utility, is to choose

$$\begin{aligned}\mathbb{P}^V &= \mathbb{P}^Y + \frac{\mathbb{P}^M - \mathbb{P}^Y}{\gamma} = \frac{1}{\gamma}\mathbb{P}^M + \left(1 - \frac{1}{\gamma}\right)\mathbb{P}^Y \\ &\approx \left(\frac{\mathbb{P}^M}{\mathbb{P}^Y}\right)^{\frac{1}{\gamma}}\mathbb{P}^Y,\end{aligned} \tag{17}$$

where $\gamma > 1$ serves as a risk-aversion parameter. From the likelihood perspective, this transformation amounts to a rescaling of the log-likelihood ratio (and hence of the profit-and-loss distribution) by a factor $1/\gamma$. As a result, extreme gains and losses are compressed, reducing drawdowns while sacrificing expected logarithmic growth.

For illustration, suppose $\mathbb{P}^M = (0.6, 0.3, 0.1)$ and $\mathbb{P}^Y = (0.5, 0.3, 0.2)$. Label the last outcome 'Loss,' which would ordinarily lead to a 50% drop in wealth. Setting $\gamma = 5$, the measure adjusts to

$$\begin{aligned}\mathbb{P}^V &= \mathbb{P}^Y + \frac{\mathbb{P}^M - \mathbb{P}^Y}{5} = (0.5, 0.3, 0.2) + \frac{1}{5}\cdot(0.1, 0, -0.1) \\ &= (0.52, 0.3, 0.18),\end{aligned}$$

yielding a new payoff vector:

$$(V_Y(1, \text{Win}), V_Y(1, \text{Draw}), V_Y(1, \text{Loss})) = (1040, 1000, 900).$$

Here, the drawdown decreases from 50% to 10%. Effectively, γ acts as a discount factor on the difference $\mathbb{P}^M - \mathbb{P}^Y$, so its preferred value chosen by the agent depends on how much the two measures disagree.

A central conclusion of this study is that the communicated payoff V_Y is *always* the growth-optimal portfolio under its implied state price density $\mathbb{P}^V$. That is, regardless of the agent's underlying belief or degree of risk aversion, once the portfolio is chosen, it is log-optimal for the distribution that it reveals. Thus, the question is not whether the Kelly criterion is inherently 'good' or 'bad' (MacLean *et al.* 2011); rather, it concerns which probability distribution $\mathbb{P}^V$ the agent should adopt. Deciding $\mathbb{P}^V$ ultimately becomes a problem of statistical modeling.

In summary, the likelihood-based viewpoint reframes portfolio construction as a form of model selection. From the market's perspective, any trading position can be taken. Once chosen, its state price density follows from $\mathbb{P}^V$, so the resulting portfolio is log-optimal under that measure. Hence, the real challenge lies in choosing $\mathbb{P}^V$ rather than settling on a particular utility function or risk measure.

9. Conclusion

We have shown that the multi-outcome Kelly solution emerges directly as the likelihood ratio $\frac{p(\omega|M)}{p(\omega|Y)}$. This portfolio invests exactly $p(\omega|M)$ fraction of wealth on each scenario ω at odds $\frac{1}{p(\omega|Y)}$. By Theorem 5.1 and Corollary 5.2, its expected logarithmic return under $\mathbb{P}^M$ is precisely the relative entropy $D_{\mathrm{KL}}(\mathbb{P}^M \| \mathbb{P}^Y)$, and no further optimization is needed to show log-utility optimality.

In markets where $\frac{p(\omega|M)}{p(\omega|Y)}$ is not replicable, the best alternative is the state price density $\mathbb{P}^V$ that lies closest to $\mathbb{P}^M$ in relative entropy. In this way, risk aversion or market frictions 'temper' an investor's view away from $\mathbb{P}^M$. While such mixing of measures has no direct analogue in pure statistics, it is a standard concept in finance. Extensions to continuous scenarios under the normal distribution are presented in Vecer *et al.* (2025), and future research might explore other continuous-time settings.

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